

PERSPECTIVE

Understanding Phosphonic-Acid Molecules Based Hole Transport Layers in Perovskite Solar Cells

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ABSTRACT

Self-assembled monolayers (SAMs) have emerged as an important class of interfacial materials in perovskite solar cells (PSCs), enabling energy level tuning, efficient charge extraction, and improving device efficiency and stability. Among them, phosphonic-acid-based SAMs stand out due to their ability to form covalent bonds with transparent conductive oxides (TCOs), working as ultrathin, transparent, and tunable hole transport layers (HTLs). Despite the rapid progress in device development, several fundamental issues remain unclear, which may limit its broad application. In this perspective, we critically analyze the progress made by the community on several scientific questions. We first examine how SAMs are chemically bonded to TCOs, which is complicated by the difficulty in characterizing SAMs coverage on rough TCO substrates. We question the general perception of how charges are collected through SAMs layer to TCOs. And finally, the stability issues related to SAMs are discussed. Addressing these remaining challenges is necessary to bring the SAMs into commercial perovskite solar cell products.

1 | Introduction

Metal halide perovskite solar cells (PSCs) have rapidly developed as a promising photovoltaic technology, due to high power-conversion efficiencies (PCEs), low manufacturing costs, and compatibility with flexible and tandem architectures [1–3]. Among the various device configurations, the p–i–n (inverted) architecture is especially appealing due to its better stability, less photocurrent hysteresis, and easier integration with narrow bandgap subcells for tandem solar cells. However, the long-term stability of p–i–n PSCs is challenged by the degradations of perovskites at the interfaces, particularly at the light-incoming hole-selective contact.

Conventional hole transport layers (HTLs), such as poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) [4], poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA)

[5], and nickel oxide (NiO_x) [6] have played important roles in PSC development but exhibit critical limitations. PEDOT:PSS is acidic and hygroscopic, leading to interfacial degradation and poor operational durability. NiO_x, while stable on its own and optically transparent, suffers from unfavorable interfacial redox activity with the perovskite absorber. The adhesion between PTAA and perovskite is weak, which induces mechanical instability of devices. Self-assembled monolayers (SAMs) can overcome those limitations of traditional HTLs by offering tunability through molecular design. A SAM molecule is generally composed of an anchoring group, a linker, and a functional terminal group. Anchoring groups of SAMs provide strong binding to the conductive oxides (TCOs), while the charge of selective backbones and terminal groups modulate energy levels, surface wettability, and may bind strongly to perovskites through coordination. The interface can be tuned to avoid reactions between the perovskite and the SAM terminal groups,

making them potentially much more stable/compatible with perovskites.

A wide variety of SAMs have been developed by the community, which can be classified by their anchoring groups into carboxylic acid [7], silane [8], and phosphonic acid (PA) SAMs [9]. Carboxylic acid-containing molecules were first used together with PTAA in p-i-n structure PSCs to modify the surface energy of HTL in 2015 [10]. Silane molecules were first shown to crosslink with fullerene to form a moisture-resistant layer in 2016 [11], and brominated (3-aminopropyl)triethoxysilane was used in 2017 to bind oxides and participate in the perovskite structure with its ammonium group, yielding a solid mechanical and electrical connection [12]. Padture et al. showed that replacing the ammonium group with an iodide group connected TCO with perovskites because the iodide group incorporated into the perovskite structure [13]. Recently, phosphonic acid-based SAMs are widely used in PSCs. Initial applications of PA-SAMs in PSCs focused on electrode surface modification [14, 15]. Albrecht and co-workers pioneered the use of carbazole-based PA as hole-selective layers (HSLs) in perovskite single-junction and tandem devices [16]. By aligning energy levels and suppressing non-radiative recombination, they achieved high device efficiencies in perovskite/CIGSe tandem solar cells. Since then, SAMs-based HSLs have rapidly emerged as a powerful class of interfacial materials, driving significant advances in p-i-n PSC development.

Advancing phosphonic-acid-based SAMs toward industrial module applications remains hindered by several challenges. This perspective critically reviews and analyzes several key scientific and technical questions related to the PA-SAMs. We first analyze the anchoring chemistry between PA-SAMs and transparent TCO surfaces, focusing on how bonding modes and anchor group multiplicity influence film quality and device stability. We then evaluate the challenges of accurately characterizing SAMs coverage and uniformity on rough TCOs. We further investigate interfacial charge dynamics from perovskite to TCO and discuss PA-SAMs-related stability pathways. Finally, we identify the key challenges and opportunities for PA-SAMs in PSCs. By integrating insights from molecular design, interface chemistry, and device performance, this perspective aims to stimulate the development of next-generation SAMs for efficient and stable perovskite photovoltaics.

2 | How are PA Groups Anchored onto TCOs?

The chemical interactions between anchoring groups and the hydroxyl groups on TCO surfaces represent a key distinction of SAMs from conventional HTLs. As illustrated in Figure 1a, the strength and extent of these interfacial reactions critically influence the coverage and packing of SAMs, which ultimately affects device efficiency and operational stability. Therefore, elucidating the mechanisms by which PA groups anchor to TCO surfaces is critical for the rational design of SAMs.

The tethering aggregation and growth (T-BAG) method has become a widely adopted approach for preparing PA-based SAMs onto TCO substrates [11–13]. This process involves: (1) deposition of a thin phosphonic acid film onto the oxide surface, which

initially leads to secondary bonding (hydrogen bonding or van der Waals interactions) between the PA headgroups and the TCO substrates, forming a weak physisorbed layer. It should be noted that SAM molecules can aggregate with each other driven by intermolecular forces, such as hydrogen bonding among PA groups [17, 18]; (2) thermal annealing, which drives dehydrative condensation reaction between hydroxyl groups from TCO and PAs to form covalent P—O—M bonds; and (3) rinsing to remove physisorbed species, leaving behind a chemically anchored monolayer [19, 20]. In this formation process, both physical and chemical absorption occur between PAs and TCO. The long-term stability of the SAMs relies predominantly on chemical interactions, that is covalent P—O—M bond formation [21–23]. In addition to covalent bonding, PA molecules may also form hydrogen bonds via their P—OH or P=O groups, either with adjacent PA units or with surface hydroxyl/oxygen groups. These interactions, however, are relatively weak, making these SAMs prone to desorb during washing or under subsequent light/thermal stress. For this reason, it is necessary to achieve stable SAMs to increase chemical bonding while decreasing weak physical adsorption. There are a few notable advances in increasing hydroxyl group density at the TCO surface to enhance SAM coverage. Han and co-workers [24] employed atomic layer deposition (ALD) to modify TCO surfaces, thereby creating substrates fully terminated with hydroxyl groups that provide well-defined anchoring sites for PA-SAMs. An alternative strategy involves surface etching of ITO using acids, which exposes fresh oxide layers enriched with hydroxyl groups [25, 26].

In addition to T-BAG, PA-SAMs can also be deposited on TCOs by solution-based self-assembly, in which the substrates are immersed in a PA solution for an extended time to allow spontaneous adsorption [28]. However, this method may not improve ordering for PA-SAMs. Compared with T-BAG, it is more time-consuming and requires a stable, nonaggregating PA-SAM solution, although it may offer higher surface coverage, which still requires experimental validation. Thermal evaporation, in principle, can form a uniform PA-SAM layer, and devices prepared by this method show efficiencies comparable to those from solution processing [29]. Nevertheless, the random molecular orientation during evaporation means that not all PA groups contact and anchor to the ITO surface, consistent with the smaller water contact angles reported by Farag et al. [29]. It also remains unclear whether PA-SAMs, particularly larger molecules, decompose at evaporation temperature.

PA molecules can adopt diverse adsorption geometries on metal-oxide surfaces, determined by the number and arrangement of their oxygen atoms and density of hydroxyl groups on TCO. As illustrated in Figure 1b, PA molecules possess three acidic oxygen atoms, which permit covalent bonding to metal-oxide surfaces through mono-, bi-, or tridentate coordination modes [24, 30, 31]. These coordination modes can occur in two distinct ways: as chelating interactions, in which two or three oxygen atoms coordinate to the same metal center, or as bridging interactions, in which each oxygen atom binds to a different metal center. The combination of these coordination and binding fashions gives rise to a rich diversity of surface-PA geometries. However, which binding mode is most stable remains unclear. Some experimental observations report that monodentate headgroups are unstable,

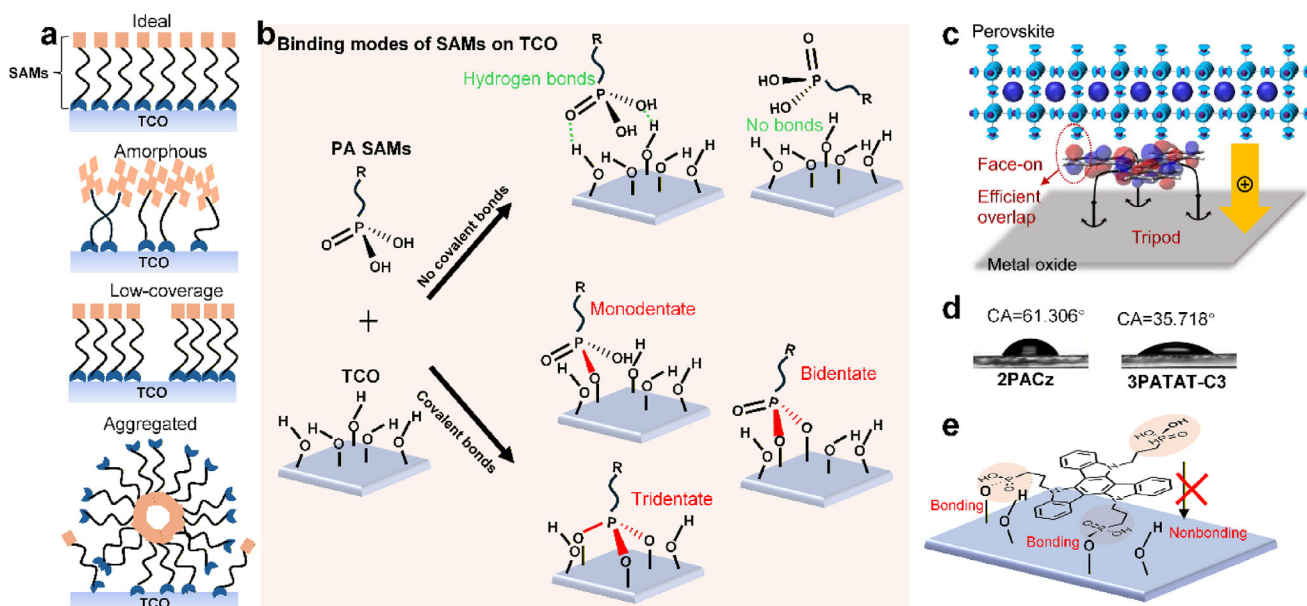


FIGURE 1 | Phosphonic-acid SAMs anchoring onto TCOs. (a) SAMs morphology, including ideal, amorphous, low-coverage, and aggregated SAM morphologies. (b) Binding modes of phosphonic-acid SAMs onto TCO surfaces. (c) The structure and experimentally determined preferential orientation of tetra-anchored triazatruxene-based SAMs [27]. Copyright 2023, American Chemical Society. (d) Water contact angles of 3PATAT-C3 and 2PACz SAMs on ITO. (e) Steric constraints force 3PATAT-C3 into an upright orientation on ITO, with some PA groups remaining nonbonding.

as they can desorb or be displaced by competing ligands in solution, leading to poor film integrity under operational stress [32, 33]. In contrast, SAMs bonding to the TCOs through two or three anchoring points can achieve more robust TCOs–SAMs interfaces [34]. However, some theoretical studies show that the most stable adsorption modes may involve a monodentate mode, in which the $p = O$ group coordinates to the TCO while the other two $P-OH$ groups bind to TCO surfaces through hydrogen bonds [24]. These seemingly contradictory findings highlight the complexity of bonding modes between PA and TCOs and the need for quantitative experimental validation.

One straightforward idea of enforcing the bonding of PA-SAMs to TCOs is to introduce multiple PA groups onto each SAM molecule, or multipodal SAMs [35]. A series of star-shaped triazatruxene derivatives were synthesized and applied in perovskite solar cells, containing one to three phosphonic-acid tethers [27]. Nonetheless, the actual benefits of multipodal SAMs have not been demonstrated yet by device performance [27]. It is unclear whether the multiple PA groups can be anchored to TCOs as designed (Figure 1c), because this would need hydroxyl groups on the TCO surface to precisely match the location of the PA groups. A tri-tethered compound 3PATAT-C3 preferentially adopts a face-on geometry on TCO surfaces, if all PAs are anchored to TCO, which would result in a very hydrophobic surface comparable to that of 2PACz. Our contact-angle measurements (Figure 1d) show that 3PATAT-C3 renders the ITO surface significantly more hydrophilic ($\theta = 35.7^\circ$) compared to 2PACz ($\theta = 61.3^\circ$), indicating that not all PA groups in 3PATAT-C3 are coordinated with TCO. The partial coordination likely arises from the limited areal density of surface $M-OH$ sites on TCOs.

In addition to the finite hydroxyl groups on TCOs, the steric constraints (Figure 1e) may also prevent the ideal face-on configu-

ration. Another work explores SAMs derived from saddle-shaped indolocyclooctatetraene π -cores, each with four PA anchors [36]. The concave π -framework effectively orients two of the four tethers “upward,” creating a bifunctional surface: a portion of the anchors binds the TCO, while the unbound tethers face outward to engage incoming perovskite precursors. This orientation is consistent with the dramatically improved surface wetting. However, the outward-facing PA groups may react with the perovskites at the interface, thereby reducing device stability. This highlights the dual role of unbound anchor groups: beneficial for wetting during perovskite film deposition, but detrimental for the long-term stability of devices.

To date, no single characterization technique has been able to unambiguously determine the bonding mode of PA-SAMs on TCOs and the fraction of $P-OH$ groups in PA-SAMs that form covalent $P-O-M$ bonds. TOF-SIMS can provide the signals of $M-O-P$ bonding PA-SAMs on TCO, but it cannot resolve the detailed binding modes [37]. Infrared spectroscopy (PM-IRRAS, FTIR) and high-field NMR can offer insights into binding configurations according to the signal of $P=O$, $P-O-H$, and $P-O-M$ at the SAM–TCO interfaces. However, these techniques still suffer from low signal-to-noise ratios, and the information remains qualitative rather than quantitative. Consequently, the absence of precise quantitative methods for probing SAMs bonding modes continues to hinder building correlations between bonding modes and interfacial stability.

3 | Challenges in Characterizing SAMs Coverage on Rough TCO Substrates

It is important to determine substrate coverage by PA-SAMs to understand their impact on efficiency and stability of devices.

Accurate quantification of coverage remains a significant challenge, particularly on rough TCO substrates. For example, some FTO substrates can display peak-to-valley topographical variations of up to ~200 nm [38]. In contrast, the thickness of SAMs is typically less than 2 nm, resulting in weak signal intensity and high sensitivity to surface roughness. This large mismatch in scale introduces significant interference and uncertainty in characterizing SAM surface coverage on TCOs using conventional techniques such as X-ray photoelectron spectroscopy (XPS), and cyclic voltammetry (CV).

On atomically flat substrates (like polished Si), XPS can provide semi-quantitative coverage by comparing the substrate signal before and after SAM deposition using a partial-coverage (island) model. The SAMs coverage (ϕ) equation [41]:

$$\phi = \frac{d_{ML}}{d_{theory}} \times \frac{n_{sub}}{n_{theory}}$$

where n_{sub} is the surface atom density of the substrate, while n_{theory} is the theoretical maximum surface density of molecules in a close-packed monolayer, calculated from the van der Waals dimensions of the SAM molecule, with the widest cross-section defining the packing limit [42]. The d_{ML} and d_{theory} represent the effective monolayer thickness determined from XPS and the theoretical thickness of a fully packed monolayer, respectively. d_{ML} can be calculated by the following equation, which accounts for attenuation of the signal of the underlying substrate by the SAM in an angle-dependent manner (θ):

$$d_{ML} = \lambda \cos \theta \ln \left(1 + \frac{I_C / (\sigma_C \lambda_C)}{I_{Si} / (\sigma_{Si} \lambda_{Si})} \right)$$

where I_x represents the integrated XPS peak intensities for element XX, one of which is unique to the underlying substrate and the other to the SAM. $\sigma_{X,i}$ is the relative sensitivity factor for a core electron in a defined orbital (i) of the element. λ_x are the inelastic mean free path (IMFP) of the photoelectrons [43, 44]. However, for rough substrates such as ITO or FTO, surface topography modifies the effective escape depth of photoelectrons and introduces topographic shadowing. As illustrated in Figure 2a [45], electrons travel a shorter path (L1) before exiting into the vacuum on smooth substrates. But for the rough substrates, electrons emitted from tilted facets must traverse a longer effective thickness (L2), resulting in additional attenuation. When analyzed with flat-surface models, leading to an apparent overestimation of monolayer thickness and coverage. In addition, photoelectrons from protrusions experience shorter escape paths and thus dominate the detected signal, whereas those from valleys or sidewalls are more strongly attenuated or partially blocked [46]. Consequently, recessed regions are under-represented and protrusions are over-represented, so XPS on rough TCO substrates reports only an apparent coverage rather than the true area fraction.

Cyclic voltammetry can assess SAM coverage and ordering on the solvent-exposed surface area on planar and rough electrodes, with accurate measurements occurring under experimental conditions that maximize surface wetting. For SAMs containing a redox-active molecule, the current passed in the oxidative and reductive

sweeps to the appended species is related to surface coverage (Γ , molecules/cm²):

$$\Gamma = \frac{Q}{nFA}$$

where Q is the charge passed, n is the number of electrodes needed to reduce or oxidize the surface-appended species, F is Faraday's constant, and A is the area of the electrode. By measuring CVs over a range of scan rates, the datasets can be used to confirm that the species are surface-bound, with the Randles-Sevcik relationship, and if the SAMs are homogeneous. Substrate roughness and porosity can perturb the electrochemical behavior of the SAM, leading to broadened or asymmetric peaks, increased peak-to-peak separations (ΔE_p), shifts in redox potential, and nonlinear scan-rate dependencies (Figure 2b) [47]. These undesirable features in the CV have been observed in many SAMs and, through detailed studies, attributed to phase segregation of molecules in mixed monolayers, lateral repulsions, or poor packing efficiency. Careful evaluation of peak shape at different scan rates can result in discrepancies in surface coverage, with slower scan rates revealing populations of surface-bound molecules with slow electron transfer rates that would go uncounted under standard experimental conditions (e.g., 100 mV/s) [48]. Nonidealities in the SAM structure can introduce significant errors in estimating surface coverage; these studies also require molecules with redox groups whose bulkiness could influence self-assembly and packing structures. CVs can also assess surface coverage of any SAM indirectly, alleviating the need to synthesize new molecules. Indirect measures do not provide surface coverage measurements but rather determine if the SAM forms an insulating layer free of pinholes. In these experiments, a redox reporter is added to the electrolyte solution (e.g., ferricyanide or ferrocene) and CVs are collected for a modified and unmodified substrate [49]. Well-packed SAMs will increase the ΔE_p values and decrease the charge passed compared to an unmodified surface, with well-insulating films blocking all electron transfer.

Neither XPS nor CV can provide the spatially resolved distribution needed to understand the correlation of coverage with perovskite morphology and related degradation. Atomic force microscopy-infrared spectroscopy (AFM-IR) has emerged as a powerful tool for mapping molecular distribution. As illustrated in Figure 2c, infrared absorption by the sample produces localized thermal expansion, which is measured by the AFM probe. When combined with resonant excitation, this strategy enables nanoscale IR spectroscopy and chemical imaging of ultrathin films, down to ~5 nm in the z-direction, including biological membranes and single monolayers [50]. Nevertheless, the inherently weak vibrational cross-sections of SAMs make their signals difficult to detect with conventional AFM-IR. Reliable AFM-IR measurements are restricted to SAMs that exhibit strong, well-separated IR-active modes outside the dominant substrate signal. Generally, a high-frequency measurement may derive more information from the top surface, which might also introduce noise (Figure 2d). In addition, the irregular surface topography of TCOs introduces interference to chemical information, a challenge for all AFM types of measurement. Taken together, these challenges highlight that AFM-IR, while uniquely powerful for nanoscale vibrational mapping, needs to be carefully conducted.

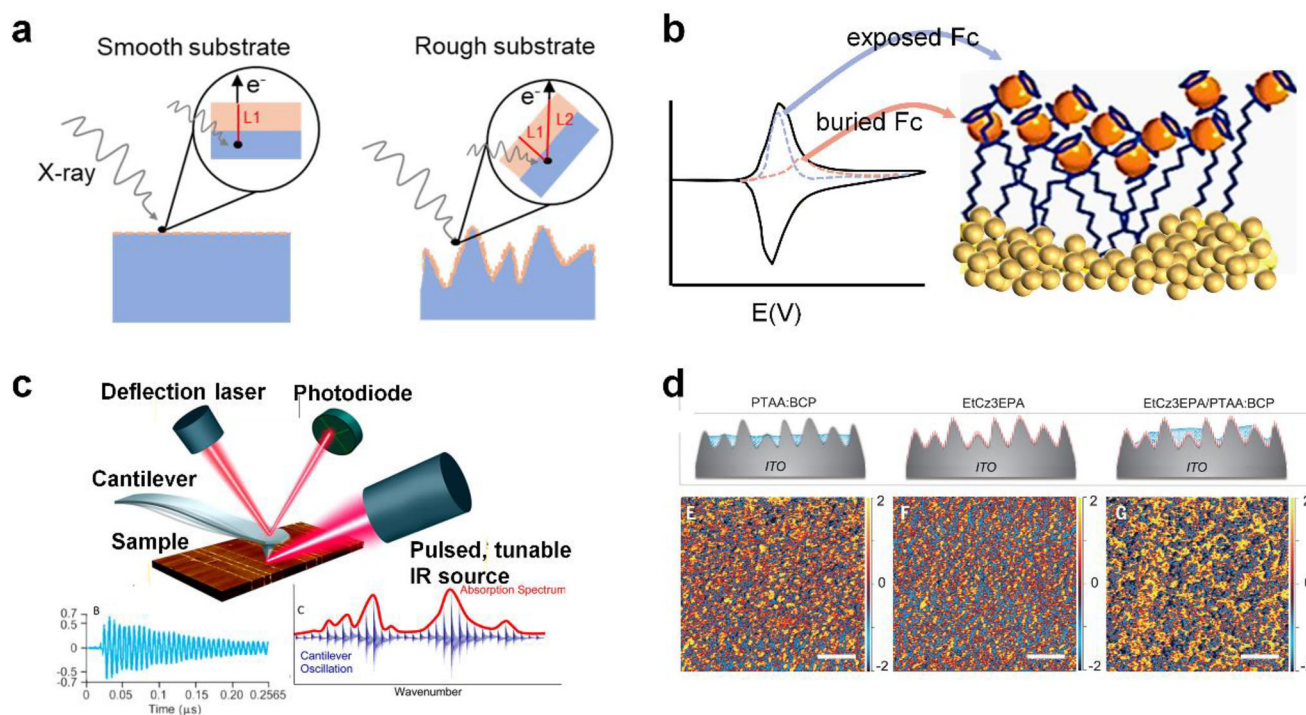


FIGURE 2 | Impact of TCO roughness on SAM coverage characterization. (a) Schematic illustration of the interference of substrate (blue) roughness on SAMs (orange) signals. The substrate surface roughness with peak-to-valley variations of $\sim 20\text{--}200$ nm induces interference in photoelectron emission, while ultrathin SAMs (< 2 nm) further complicate accurate coverage quantification. L1 and L2 represent the effective photoelectron escape distances on smooth and rough substrates, respectively, with roughness increasing the escape path ($L2 > L1$). (b) Effect of surface roughness on the (scan rate-dependent) shape of CVs. SAMs can develop structural disorder on rough substrates, such as phase segregation, lateral repulsion, or loose molecular packing (right), leading to broadened or asymmetric CV peaks that can be overlooked at fast scan rates (left) [39] copyright 2015, American Chemical Society. (c) Schematic of AFM-IR system enabling sub-100 nm resolution imaging through an IR-activated cantilever response [40] Copyright 2017, American Chemical Society. (d) AFM-IR mapping of CH_3 bending vibrations ($1012\text{--}1028\text{ cm}^{-1}$) reveals spatial inhomogeneity in SAMs [3]. Copyright 2024, The American Association for the Advancement of Science.

4 | How are Charges Collected Through SAMs Layer to TCOs?

Hole extraction from the perovskite absorber to the TCO electrode is crucial for both device efficiency and stability. For PTAA or NiO_x , a thickness of ~ 10 nanometers or larger is common to ensure good coverage while minimizing series resistance. SAMs are generally much thinner ($1\text{--}2$ nm), however, PSCs employing most SAMs often exhibit reduced charge collection efficiency, mainly due to the intrinsically lower conductivity of SAMs. Despite that most of the terminal functional groups are designed to be semiconducting, such as triphenyl amine and carbazole groups, the PA anchoring group and the linker are mostly insulating. This raises a fundamental question: do SAMs operate as a tunneling layer, or do they function similarly to insulating oxides like Al_2O_3 , which showed great success in perovskite solar cells (Figure 3) [51]. The long carrier diffusion lengths in perovskites make it possible that photogenerated charges can diffuse around the island of PAs and be collected by the TCO directly. Under this context, the function of PAs is to modify the overall work function of TCO and provide possible passivation or mechanical interaction with perovskites, very different from traditional hole transport layers.

Some experimental evidence favors a tunneling mechanism. Time-resolved photoluminescence measurements reveal that the

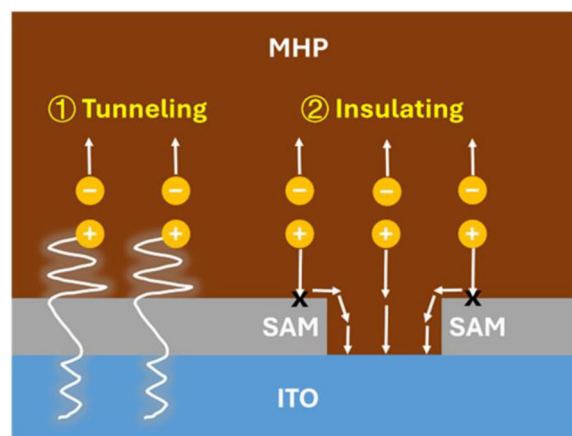


FIGURE 3 | Schematic illustration of SAMs acting as (1) tunneling contacts and (2) diffusion barrier in PSCs.

hole extraction process in PSCs using SAMs is comparable to or even faster (depending on the molecular structure) than that in PSCs using PTAA [51]. However, the non-continuity of both PA molecules and PTAA should be considered, since most perovskites may still directly contact with TCO, causing faster charge transfer. Linker thickness-dependent studies conducted by Levine et al. further support the tunneling mechanism. They

found that the hole transfer rate is significantly reduced with increasing length of the alkyl spacer in nPACz ($n = 2,4,6$) SAMs [52]. However, this assumes a complete monolayer of these molecules assembling with the same tilting angle as found on flat substrates, and a conserved perovskite morphology grown on the different substrates. More study is needed to determine whether the conductivity of the PA and other SAMs can sustain the photocurrent density generated under 1 sun illumination while consuming negligible photovoltage.

Since it is unlikely to change the conductivity of the anchor group, some efforts were devoted to modifying the conductivity of the linkers. Qu et al. [52] replaced the flexible alkyl linker with a conjugated phenylene linker in Me-4PACz. Zhao et al. [53] designed a series of SAMs featuring increasingly extended π -conjugation structures while maintaining a fixed spacer length and anchoring group. They found that as the π -conjugation of the SAMs core structure was extended, leading to hole extraction being enhanced. However, there are other cases showing conjugation does not necessarily enhance hole extraction, because the linker group also determines the packing of the PA molecules. Changing from the flexible alkane chain to rigid planar conjugated structure may make the carbazole group change from edge on to face orientation [54], and thus increasing the PA molecule layer thickness.

4.1 | Other Stability Issues Related to SAMs

As SAMs reside at the interface between TCOs and the perovskite absorber, the long-term stability of SAM-based devices is influenced by both the TCO/SAM and SAM/perovskite interfaces. At the TCO/SAM interface, instability often arises from SAM desorption. In addition to the desorption of SAMs from TCOs and the detachment of SAMs from perovskites, the dangling acid groups of PA-SAMs may also participate in unwanted electrochemical reactions with perovskites. We recently showed that the acidic protons of PAs can accelerate the oxidation of iodide, decompose formamidine, and reduce lead ions, and these reactions are accelerated at high temperature and exposure to ultraviolet light [54]. So, PA molecules weakly bonded to ITO, or exposing the PA groups due to either head-to-head packing or nonideal multipodal coordination would react with perovskites [55]. Another instability issue comes from the unreacted hydroxyl groups on the TCO. Increasing hydroxyl group density on the TCO can enhance the number of reactive sites for anchoring of the PAs; however, it could be a double-edged sword, introducing more unreacted —OH groups if the SAM coating process cannot consume all of them. Even covalent P—O—M linkages formed by phosphonic acid SAMs are not immune to degradation. Under humid conditions, hydrolytic cleavage of the P—O bond generates free hydroxyl groups on phosphorus, which is the reverse process of dehydration, causing desorption of the SAMs. This forces SAM processes to be done in moisture-free or low moisture environments. Therefore, enhancing the bonding energy of SAMs with TCO would make it more difficult to hydrate and thus enable processing in air. UV light during long-term operation may also break down the covalent bonds, whose computed bonding energy between PA-SAMs with TCO is around or smaller than 3 eV [3]. However, these hypotheses still need rigorous verification. To mitigate these challenges, Jiang et al. [56] developed a cross-

linkable strategy to form robust covalent bonds of hole-selective SAMs on TCOs under external stress. Specifically, they incorporated azide-containing moieties into the SAM, which undergo thermally activated cross-linking to form a dense, oriented SAMs network with improved thermal stability. Similarly polymerized SAMs with multiple PA-groups should also enhance the UV stability of PA-SAMs [57]. It remains to see whether all the PAs in the polymeric SAMs would react with TCO rather than FA cations in perovskites, though the chemical design may overcome this limitation.

While interactions of PA-SAMs may strengthen the adhesion of perovskites to TCOs, the excessive stress of this configuration may induce faster ion migration and phase segregation/transition. Furthermore, direct contact between TCOs and perovskite—arising from incomplete or low SAM coverage—can exacerbate instability. FA-based perovskites are prone to decomposition under heat or humidity, generating acidic species [58]. The basic nature of metal-oxide TCOs can facilitate this degradation pathway. Although no instability is evident over several thousand hours of operation, issues may emerge on longer timescales [59], which need further study.

5 | Conclusions and Outlook

In summary, PA-SAMs have emerged as highly promising interlayers for bridging TCOs and perovskite absorbers, but their further successful deployment requires resolving several fundamental challenges. At the most basic level, the bonding strength and modes of phosphonic acid anchoring onto TCOs remain insufficiently understood. These bonding properties are governed not only by the chemistry of the anchoring group and packing of the PA-SAMs, but also by the distribution of hydroxyl groups on the TCO. Surface engineering (e.g., ALD-generated —OH terminations, TCO chemical etching) can improve —OH density, but the bonding modes remain unknown. Second, incomplete coordination of all phosphonate headgroups often leaves unbound anchors exposed at the interface, causing deleterious chemical interactions with the perovskite absorber that undermine long-term device stability.

Future ligand designs must therefore strike a careful balance—minimizing steric bulk through regulating the condensation reaction to ensure full bonding of all anchoring groups—essential for multipodal SAMs to achieve both high efficiency and long-term stability in perovskite photovoltaics. Equally critical is the reliable quantification of SAM coverage and uniformity, which remains challenging because ultrathin films produce weak signals that are further complicated by substrate roughness. Future research will require the development of advanced characterization techniques capable of enhancing SAM-specific signals while suppressing substrate interference. Integrating data-driven approaches, such as machine-learning-assisted spectral analysis and image processing, may provide powerful tools to extract meaningful coverage information from noisy or spatially heterogeneous datasets.

Charge transport across the SAMs interface represents another frontier. This will inspire SAMs design: future molecular engineering should focus on minimizing tunneling barriers through optimized energy-level alignment, extended π -conjugation, and

so on. Although PA-SAM is prevalent now, other types of anchoring molecules should be equally important in principle. Some stability issues of SAMs in PSCs should be considered carefully. Instability often originates from the SAM desorption and residual surface hydroxyl groups, of which reactive dangling acid groups can further damage perovskites through chemical reaction.

Finally, the scalable coating of SAMs remains to be a challenge. The T-BAG method imposes some technical difficulty in manufacturing, particularly the washing step. Washing-free SAMs would be preferred to bring SAMs into the products.

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Conflicts of Interest

The authors declare no conflicts of interest.

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